Kenilkumar Surani

Mechanical Design Engineer | SolidWorks CSWP

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Professional Summary

Mechanical Design Engineer and M.Sc. Interdisciplinary Engineering student in Germany with strong expertise in 3D CAD modeling, machine design, and manufacturing documentation. Certified SolidWorks Professional (CSWP) experienced in designing complex mechanical assemblies, industrial machinery, and EV drivetrain systems. Skilled in SolidWorks, GD&T, tolerance analysis, and mechanical system design with hands-on project experience in automotive and mechanical engineering applications.

Technical Skills

- CAD & Design Software
- SolidWorks (CSWP), Solid Edge, Autodesk Inventor, AutoCAD, KeyShot
- Engineering & Analysis
- GD&T, Tolerance Stack-up Analysis, Finite Element Analysis (Ansys Workbench)
- Manufacturing & Documentation
- Design for Manufacturing (DFM), Design for Assembly (DFA), BOM Creation, Mechanical Assembly Design, Technical Drawings, Exploded Views

Professional Experience

SolidWorks Drafter – Radhe Krishna Enterprise, Surat, India (05/2021 – 11/2021)

- Designed precision mechanical components for diamond polishing machines using SolidWorks, focusing on durability and manufacturability.
- Created detailed 2D manufacturing drawings with GD&T to ensure accurate machining and assembly.
- Collaborated with production engineers to refine designs and improve mechanical reliability.
- Prepared assembly layouts and technical documentation to support manufacturing processes.

Mechanical Designer (Remote) – Punneli Technologies, Dublin, Ireland (10/2020 – 01/2021)

- Designed and modeled assemblies for an automatic plastic tube filling and sealing machine using SolidWorks.
- Developed full machine assemblies, sub-assemblies, and detailed part drawings for manufacturing.
- Generated BOMs, exploded views, and technical documentation for fabrication and assembly.
- Collaborated with the company director to refine mechanical design based on production requirements.

Key Engineering Projects

Manual Variable Transmission (MVT) for Electric Vehicle | 2022

- Designed a multi-ratio transmission system for electric bikes to improve efficiency, acceleration, and hill-climbing capability.
- Modeled the full transmission assembly in SolidWorks including cone pulleys, V-belt system, and power screw actuators.
- Produced engineering drawings, exploded views, and BOM for prototype manufacturing.
- Integrated the transmission system into an electric bike prototype and conducted mechanical testing to evaluate performance and efficiency improvements.

Automatic Tube Filling & Sealing Machine | 2021

- Designed mechanical assemblies for an industrial tube filling machine used in packaging production.
- Modeled machine components and assemblies using SolidWorks.
- Prepared manufacturing drawings and assembly documentation for fabrication.
- Optimized mechanical layout to improve component integration and assembly efficiency.

Electric Treadmill Tricycle | 2021

- Designed a hybrid human-powered and electric tricycle system to promote ergonomic transportation.
- Modeled frame structure, drivetrain system, and treadmill integration using SolidWorks.
- Developed assembly documentation focused on energy efficiency and ergonomic performance.

Education

M.Sc. Interdisciplinary Engineering – Hochschule Trier, Germany (09/2023 – Present)

- Focus Areas: Vehicle Safety, Crash Analysis, Finite Element Method (FEM), CAD, Electronic Engine Management, SAP ERP Production Management, International Management
- Project: Conducted crash simulation of a vehicle B-pillar during side impact using Ansys Workbench.

B.Tech. Automobile Engineering – Uka Tarsadia University, India (07/2018 – 01/2023)

- Core Subjects: Automotive CAD, Vehicle Dynamics, Auto Chassis & Body Engineering, Fluid Mechanics, Thermodynamics, Automotive Aerodynamics
- Thesis: Manual Variable Transmission (MVT) for Electric Vehicle – designed full transmission assembly in SolidWorks.

Certifications

Certified SolidWorks Professional (CSWP) – Mechanical Design, Drawing Tools, Sheet Metal, Mold Making, Surfacing

MITx edX – Introduction to Aerospace Engineering

edX – Electric Cars: Business & Technology

Languages

English – C1 (Fluent) / German – A2 / Hindi – Native